

Chapter 3

Percents and Their Applications

Math for Business and Finance: an Algebraic Approach, 3rd Edition
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Learning Unit Objectives

LU 3-1: Conversions

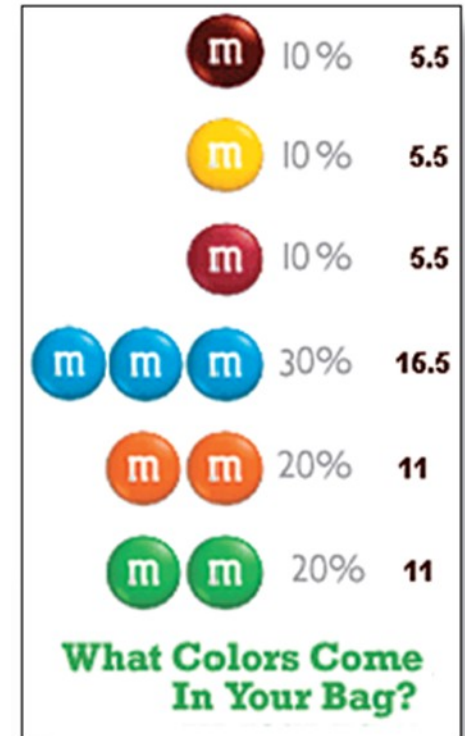
1. Convert decimals to percents (including rounding percents), percents to decimals, and fractions to percents.
2. Convert percents to fractions.

LU 3-2: Application of Percents—Portion Formula.

1. List and define the key elements of the portion formula.
2. Solve for one unknown of the portion formula when the other two key elements are given.
3. Calculate the rate of percent increases and decreases.

Table 3.1 - Bag of M&M'S

color	Fraction	Decimal (hundredth)	Percent (hundredth)
Yellow	$\frac{18}{55}$	.33	32.73%
Red	$\frac{10}{55}$	.18	18.18%
Blue	$\frac{9}{55}$	.16	16.36%
Orange	$\frac{7}{55}$	.13	12.73%
Brown	$\frac{6}{55}$	.11	10.91%
Green	$\frac{5}{55}$	.09	9.09%
Total	$\frac{55}{55} = 1$	1.00	100.00%



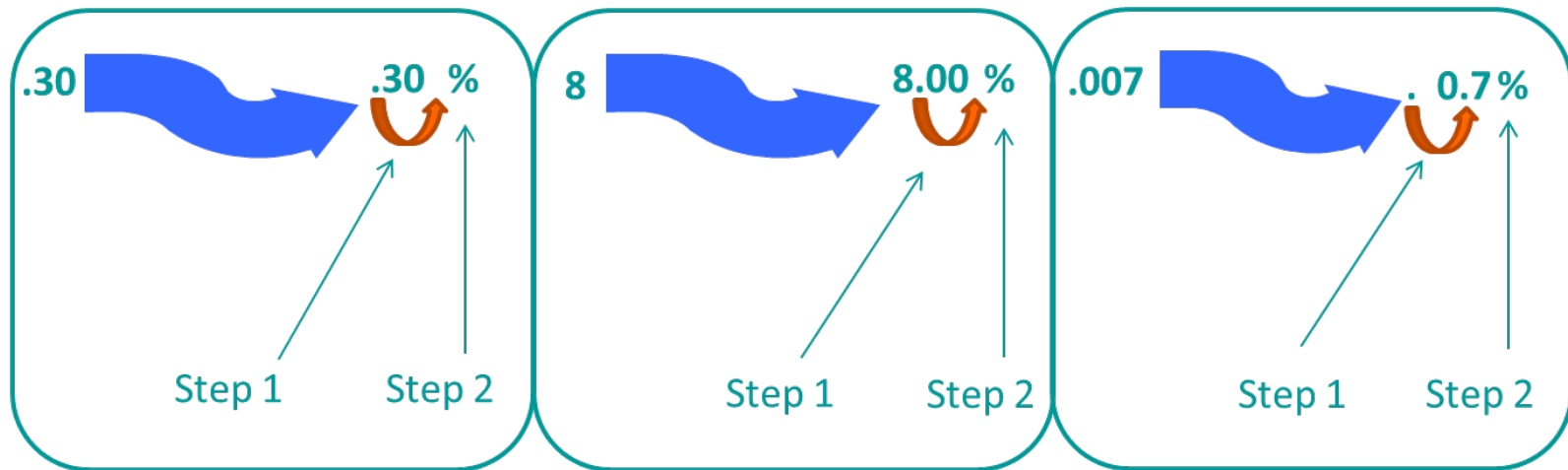
Information adapted from <http://us.mms.com/us/about/products/milkchocolate/>

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Converting Decimals to Percents

2 Steps

1. Move decimal point **two** places to the right. You are multiplying by 100. If necessary, add zeros. This rule is also used for whole numbers and mixed decimals.
2. Add a percent symbol at the end of the number.



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Rounding Percents

3 Steps

1. When you convert from a fraction or decimal, be sure your answer is in percent form before rounding.
2. Identify the specific digit. If the digit to the right of the identified digit is 5 or greater, round up the identified digit.
3. Delete digits to the right of the identified digit.

Example

Round $\frac{18}{55}$ to the nearest hundredth percent:

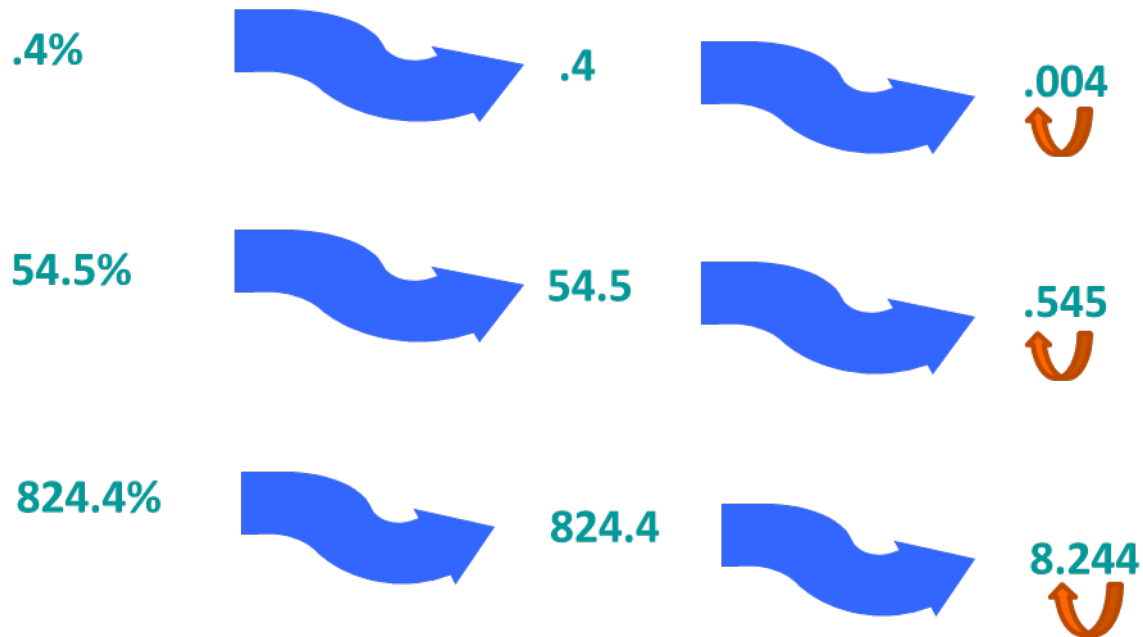


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Converting Percents to Decimals

2 Steps

1. Drop the percent symbol.
2. Move decimal point **two** places to the left. You are dividing by 100. If necessary, add zeros.



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Converting Fractional Percents To Decimals

3 Steps

1. Convert a single fractional percent to its decimal equivalent by dividing the numerator by the denominator. If necessary, round the answer.
2. If a fractional percent is combined with a whole number (mixed fractional percent), convert the fractional percent first. Then combine the whole number and the fractional percent.
3. Drop the percent symbol; move the decimal point **two** places to the left (this divides the number by 100).

$$\frac{1}{4}\% \rightarrow \frac{1}{4} = .25\% \rightarrow .0025$$

$$7\frac{3}{4}\% \rightarrow 7\frac{3}{4} = 7.75\% \rightarrow .0775$$

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Converting Fractions to Percents

2 Steps

1. Divide the numerator by the denominator to convert the fraction to a decimal.
2. Move decimal point **two** places to the right; add the percent symbol.

The diagram illustrates the conversion of the fraction $\frac{3}{4}$ to the percent 75%. It features a blue wavy arrow pointing from left to right. Inside the arrow, the text $3 / 4 =$ is followed by a space and then the decimal $.75$. To the left of the arrow's start is the fraction $\frac{3}{4}$ in teal. To the right of the arrow's end is the percent 75% in teal, with a small orange curved arrow pointing upwards towards it.

$$\frac{3}{4} \rightarrow 3 / 4 = .75 \rightarrow 75\%$$

The diagram illustrates the conversion of the fraction $\frac{1}{5}$ to the percent 20%. It features a blue wavy arrow pointing from left to right. Inside the arrow, the text $1 / 5 =$ is followed by a space and then the decimal $.20$. To the left of the arrow's start is the fraction $\frac{1}{5}$ in teal. To the right of the arrow's end is the percent 20% in teal, with a small orange curved arrow pointing upwards towards it.

$$\frac{1}{5} \rightarrow 1 / 5 = .20 \rightarrow 20\%$$

Converting a Whole Percent (or a Fractional Percent) to a Fraction

3 Steps

1. Drop the percent symbol.
2. Multiply the number by $\frac{1}{100}$.
3. Reduce to lowest terms.

156%

156

$$156 \times \frac{1}{100} =$$

$$\frac{156}{100} =$$

$$1 \frac{56}{100} =$$

$$1 \frac{14}{25}$$

$\frac{1}{8}$ %

$\frac{1}{8}$

$$\frac{1}{8} \times \frac{1}{100} =$$

$$\frac{1}{800}$$

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Converting a Mixed or Decimal Percent to a Fraction

4 Steps

1. Drop the percent symbol.
2. Change the mixed percent to an improper fraction.
3. Multiply the number by $\frac{1}{100}$.
4. Reduce to lowest terms.

Note: If you have a mixed or decimal percent, change the decimal portion to its fractional equivalent and continue with Steps 1 to 4.

The diagram illustrates the conversion process for two types of percents. For a mixed percent, $12\frac{1}{2}\%$, a blue arrow points from the mixed number to the calculation $\frac{25}{2} \times \frac{1}{100} = \frac{25}{200}$, which then simplifies to $\frac{1}{8}$. For a decimal percent, 12.5% , a blue arrow points from the decimal to the mixed number $12\frac{1}{2}\%$, and another blue arrow points from the mixed number to the same calculation and simplification steps: $\frac{25}{2} \times \frac{1}{100} = \frac{25}{200} = \frac{1}{8}$.

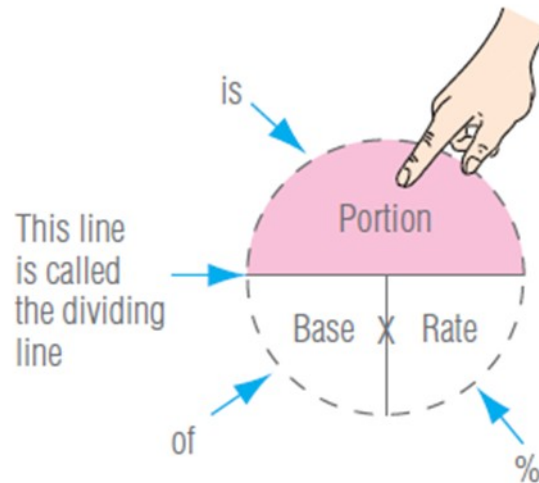
$$12\frac{1}{2}\% \rightarrow \frac{25}{2} \times \frac{1}{100} = \frac{25}{200} = \frac{1}{8}$$
$$12.5\% = 12\frac{1}{2}\% \rightarrow \frac{25}{2} \times \frac{1}{100} = \frac{25}{200} = \frac{1}{8}$$

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Solving Percents with the Portion Formula

- When solving problems involving portion, base, or rate, you must be given two of these elements.
- You solve for the third.

$$\text{Portion (P)} = \text{Base (B)} \times \text{Rate (R)}$$

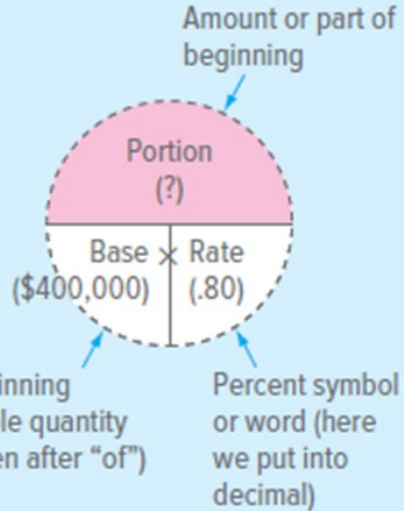


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Solving for Portion

Sales of Milk Chocolate M&M'S® are 80% of total M&M'S® sales. Total M&M'S® sales are \$400,000. What are the sales of Milk Chocolate M&M'S®?

$$P = \$400,000 \times .80 = \$320,000$$

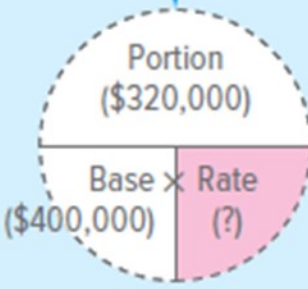
	The facts	Solving for?	Steps to take	Key points
BLUEPRINT	<i>Milk Chocolate M&M'S® sales: 80%.</i> <i>Total M&M'S® sales: \$400,000.</i>	Sales of Milk Chocolate M&M'S®.	Identify key elements. Base: \$400,000. Rate: .80. Portion: ? Portion = Base × Rate.	 Amount or part of beginning Portion (?) Base (\$400,000) × Rate (.80) Beginning whole quantity (often after "of") Percent symbol or word (here we put into decimal) Portion and rate must relate to same piece of base.

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Solving for Rate

Sales of Milk Chocolate M&M'S® are \$320,000. Total M&M'S® sales are \$400,000. What is the percent of Milk Chocolate M&M'S® sales compared to total M&M'S® sales?

$$\text{Rate} = \$320,000 / \$400,000 = 80\%$$

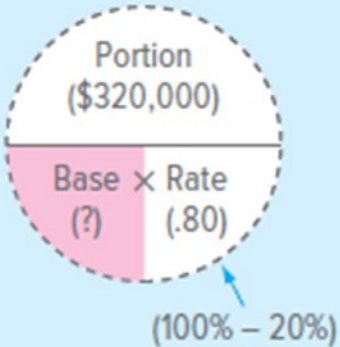
	The facts	Solving for?	Steps to take	Key points
BLUEPRINT	<i>Milk Chocolate M&M'S® sales: \$320,000.</i> <i>Total M&M'S® sales: \$400,000.</i>	Percent of Milk Chocolate M&M'S® sales to total M&M'S® sales.	Identify key elements. Base: \$400,000. Rate: ? Portion: \$320,000 $\text{Rate} = \frac{\text{Portion}}{\text{Base}}$	Since portion is less than base, the rate must be less than 100%  Portion and rate must relate to the same piece of base.

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Solving for Base

Sales of Peanut and other M&M'S® chocolate candies are 20% of total M&M'S® sales. The remaining sales are from Milk Chocolate M&M'S®, which are \$320,000. What are the total sales of all M&M'S®?

$$B = \$320000 / .80 = \$400000$$

	The facts	Solving for?	Steps to take	Key points
BLUEPRINT	<i>Peanut and other M&M'S® chocolate candies sales: 20%.</i> <i>Milk Chocolate M&M'S® sales: \$320,000.</i>	Total M&M'S® sales.	Identify key elements. Base: ? Rate: .80 (100% – 20%) Portion: \$320,000 Base = $\frac{\text{Portion}}{\text{Rate}}$	 Portion (\$320,000) and rate (.80) do relate to the same piece of base.

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Calculating Percent Increases and Decreases

2 Steps

1. Find the difference between amounts (such as price per load).
2. Using the formula $R \times B = P$, solve for R . Be sure to express your answer in percent.

Rate of Percent Increase

Sheila Leary went to her local supermarket and bought a bag of M&M'S®. The bag gave its weight as 18.40 ounces, which was 15% more than a regular 1-pound bag of M&M'S®. Sheila, who is a careful shopper, wanted to check and see if she was actually getting a 15% increase.

Steps to solving problem

1. Set up the formula.

$$\text{Rate} \times \text{Base} = \text{Portion}$$

$$R \times B = P$$

2. Calculate the rate.

$$R \text{ of } 16.00 \text{ oz.} = 2.40 \text{ oz.}$$

$$16.00R = 2.40$$

$$\text{Rate} = .15 \text{ or } 15\% \text{ increase}$$



$$\frac{\cancel{16.00}R}{\cancel{16.00}} = \frac{2.40}{16.00}$$

The new weight of the bag of M&M'S® is really 115% of the old weight:

$$16.00 \text{ oz.} = 100\%$$

$$\frac{+ 2.40}{18.40 \text{ oz.}} = \frac{+ 15}{115\% = 1.15}$$

Rate of Percent Decrease

The increase in the price of sugar caused the M&M/Mars company to decrease the weight of each 1-pound bag of M&M'S® to 12 ounces. What is the rate of percent decrease?

Steps to solving problem

1. Set up the formula.

$$\text{Rate} \times \text{Base} = \text{Portion}$$

$$R \times B = P$$

2. Calculate the rate.

$$R \text{ of } 16.00 \text{ oz.} = 4.00 \text{ oz.}$$

$$16.00R = 4.00$$

$$\frac{\cancel{16.00}R}{\cancel{16.00}} = \frac{4.00}{16.00}$$

$$R = .25 = 25\%$$

Rate = .25 or 25% decrease



The new weight of the bag of M&M'S® is really 75% of the old weight:

$$16 \text{ oz.} = 100\%$$

$$\begin{array}{r} - 4 \\ \hline 12 \text{ oz.} \end{array} = \begin{array}{r} - 25 \\ \hline 75\% \end{array}$$

Textbook Problem 3-6

Problem Statement:

Convert the following decimal to percent: *LU 3-1(1)*

6.006

Solution:

600.6% ← Add % sign



Textbook Problem 3-11

Problem Statement:

Convert the following percent to decimal: *LU 3-1(1)*

119%

Solution:

1.19 ← Drop % sign



Textbook Problem 3-15

Problem Statement:

Convert the following fraction to percent (round to the nearest tenth percent as needed): *LU 3-1(2)*

$$\frac{7}{8}$$

Solution:

$$\frac{7}{8} = .875 = 87.5\% \quad \leftarrow \text{Add \% sign}$$


Textbook Problem 3-19

Problem Statement:

Convert the following percent to a fraction and reduce to the lowest terms: *LU 3-1(2)*

$$31\frac{2}{3}\%$$

Solution:

$$31\frac{2}{3}\% = \frac{95}{3} \times \frac{1}{100} = \frac{95}{300} = \frac{19}{60}$$

Textbook Problem 3-26

Problem Statement:

Solve for the portion (round to the nearest hundredth as needed): *LU 3-2(2)*

119% of 128.9

Solution:

Portion = Rate $\times$ Base

Portion = 1.19 $\times$ 128.9

Portion = 153.39

Textbook Problem 3-34

Problem Statement:

Solve for the base (round to the nearest hundredth as needed): *LU 3-2(2)*

36 is .75% of what number?

Solution:

$$\text{Base} = \frac{\text{Portion}}{\text{Rate}}$$

$$\text{Base} = \frac{36}{.0075}$$

$$\text{Base} = 4,800$$

Textbook Problem 3-42

Problem Statement:

Solve for rate (round to the nearest tenth percent as needed): *LU 3-2(2)*

.09 is what percent of 2.25?

Solution:

$$\text{Rate} = \frac{\text{Portion}}{\text{Base}}$$

$$\text{Rate} = \frac{0.09}{2.25}$$

$$\text{Rate} = .04 = 4\%$$


End of Main Content



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